The Theory of Evolution

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Alfred Wallace was an English naturalist who independently arrived at similar conclusions to Charles Darwin concerning the theory of natural selection, often referred to as 'survival of the fittest'.

1 Study the definition of natural selection given below and rewrite it in your own words.

> **Natural selection:** A process resulting in the survival and reproductive success of organisms best adjusted to a specific environment, and that leads to the passing on of genetic qualities best suited to that environment.

Darwin had worked out most of his theories by the 1840s but was reluctant to publish them, perhaps aware of the controversy they would cause. Wallace wrote to Darwin in 1858 setting out his theory of natural selection as a mechanism for evolution, which was virtually identical to Darwin's ideas. Although Darwin suggested that the two present their work together, Wallace graciously declined. He allowed Darwin to publish his work first, on the grounds that Darwin had spent many years collecting evidence with which to support it. One viewpoint is that Darwin cheated Wallace of recognition for his work, but today we acknowledge the efforts of both men. Wallace was also spared much of the religious backlash Darwin experienced.

2 Why might some people take exception to Darwin's theories on evolution? (*Hint*: look at the quote below by Benjamin Disraeli, the British Prime Minister, in 1864.)

> 'Is man an ape or an angel? I, my lord,
> I am on the side of the angels.'



Copy the following passage on the *Theory of Evolution* and fill in the missing words:

> Parents produce more (**3**) than can survive. Not all offspring are (**4**) to their parents. The variation between individuals will make some better able to (**5**) the pressures of nature than others. The (**6**) (features) that make the individual better suited to survive will be passed on to their offspring. Because many will possess this trait, over time it will (**7**) in the population. Those without it will (**8**) or produce fewer surviving offspring.

Food chains and food pyramids

Food chains and food pyramids, respectively, show the links between 'who eats who' and the relative numbers involved.

9 Which type of species in a food chain will produce large numbers of offspring and which will produce fewer?

10 Why do you think evolution has produced the result you gave as your answer for Question 9?

Darwin was only able to explain evolution on a theoretical basis. The true field of genetics was not born until the work of Gregor Mendel, an Austrian monk and contemporary of Darwin, was made public.

11 Explain what is meant by 'theoretical'.

12 Explain what is meant by 'contemporary'.

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